

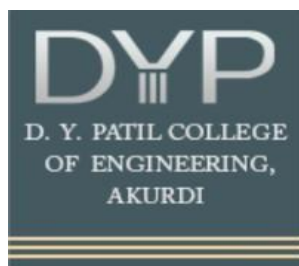
**A Case Study Report**  
*on*  
**“Online Voting System”**

Submitted  
For the partial fulfillment for the award of the Degree of  
Bachelor of Technology  
in  
Information Technology

| Roll no.   | Name Of Student  |
|------------|------------------|
| SY24INT024 | Mugdha Mane      |
| SY24INT029 | Yash Suryawanshi |
| SY24INT050 | Pradnya Patil    |
| SY24INT070 | Yashvi Gheewala  |

Under the guidance of

**Mrs. Devyani Girasey**



Department Of Information Technology

D. Y. Patil College of Engineering  
Dr D.Y Patil Educational Complex, Sector.29, Akurdi, Pradhikaran, Maharashtra, India  
**2025-2026**

## **CONTENT TO BE COVERED IN CASE STUDY:**

1. Problem definition and objectives
  2. Data collection
  3. Methodology/approach
  4. Implementation and result
  5. Conclusion
- 

### **❖ Case Study**

### **Online Voting System**

#### **1. Problem Definition & Objectives**

##### **➤ Problem Definition**

Traditional voting methods (paper ballots or EVMs) have several limitations in terms of accessibility, transparency, and efficiency.

In the existing system:

- Voters must visit polling booths physically
- Voting and counting require manual effort
- Centralized systems control the process

However, this leads to serious problems:

##### **✚ Major Issues:**

- Lack of transparency in vote counting
- Possibility of vote tampering or fraud
- Time-consuming process
- Low voter turnout due to inconvenience
- High cost of conducting elections
- Limited accessibility for remote voters

These limitations create **trust issues, financial risks, and inefficiency** in the system

## Objectives

To design and develop a **secure, efficient, and user-friendly online voting system.**

## Primary Aim

To develop a **decentralized, secure, and automated chit fund system using blockchain technology.**

### ➤ Detailed Objectives:

- Enable **remote voting from any location.**
- Ensure **secure and fair elections.**
- Maintain **voter privacy and confidentiality.**
- Prevent duplicate or fake voting.
- Automate vote counting.
- Provide real-time result updates.
- Reduce cost and time of elections.

## 2. Data Collection

### ➤ Types of Data Collected

The system collects and processes the following data:

#### Voter Data:

- Voter ID / Username
- Password or authentication details
- Voting status

#### Election Data:

- Election ID
- Candidate names and details

#### Vote Data:

- Selected candidate
- Time of voting

#### Result Data :

- Total votes per candidate
- Final Results

➤ **Sources of Data**

- **User Input:**
  - Registration
  - Voting
- **Admin input**
  - Candidate details
- **System-generated data**
  - Votes and results

➤ **Data Characteristics**

- Secure and confidential
- Accurate and reliable
- Stored digitally
- Easily accessible for authorized users

### **3. Methodology / Approach**

➤ **System Approach**

**The system follows a client-server architecture.**

**System Layers:**

**1. Frontend (User Interface)**

Web **interface** for voters and admin.

**2. Backend**

Handles logic like authentication and vote processing

**3. Database**

**Stores voter details, votes, and results**

➤ **Step-by-Step Workflow**

- 1. User registers in the system**
- 2. Admin verifies voter details**
- 3. User logs in securely**

4. **User views list of candidates**
5. **User casts vote**
6. **System records the vote**
  
7. **User cannot vote again**
8. **Votes are counted automatically**
9. **Results are displayed**

#### ► **Methodologies Used**

##### **Authentication System**

- Username/password or OTP-based login

##### **Database Management**

- Stores all voting data securely

##### **Validation Mechanism**

- Ensures one voter can vote only once

#### ► **System Processes**

- Voter Registration
- Login Authentication
- Vote Casting
- Vote Storage
- Vote Counting
- Result Display

## 4. Implementation & Results

### ► Implementation Details

#### Technologies Used:

- Frontend: HTML, CSS, JavaScript
- Backend: Java
- Database: MySQL

### ► Key Features Implemented

- Secure login system
- Easy voting interface
- One vote per user restriction
- Automatic vote counting
- Instant result generation

### ► Results Achieved

1. **Efficiency**  
Faster voting and counting process
2. **Accuracy**  
Eliminates human errors
3. **Security**  
Protects voter data
4. **Convenience**  
Vote from anywhere
5. **Transparency**  
Clear and visible results

### ► Performance Outcomes

- Reduced election time
- Increased voter participation
- Lower operational cost
- Improved reliability

## 5. Conclusion

The **Online Voting System** simplifies the election process by making it digital, secure, and efficient.

It ensures:

- Easy access for voters
- Accurate and fast results
- Reduced chances of fraud

### **Future Enhancements:**

- Biometric authentication
- Mobile application
- Blockchain integration for higher security
- Multi-language support